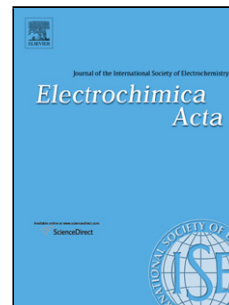


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Synthesis and properties of new carboxyborate lithium salts as electrolytes for lithium-ion batteries

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Abstract

Bis(carboxytrifluoroborate lithium) salts $[R(\text{COOBF}_3\text{Li})_2]$ with oxyethylene groups R of oligomeric molar masses $[R = \text{O}(\text{CH}_2\text{CH}_2\text{O})_n]$, where $n = 3$ or 11 , BCB3 and BCB11, respectively] were synthesized via reaction of carboxylates salts with boron fluoride. The new salts were characterized by spectroscopic analysis. The physical properties of the salts were determined by oxyethylene chain length. For $n = 3$ the salt was crystalline with $m_p = 197^\circ\text{C}$ and for $n = 11$ it showed properties of an ionic liquid at ambient temperature. Their thermal stability was at least 250°C . The values of lithium-ion transference numbers (T_+) of the solutions in polar aprotic solvents, determined by a well established steady-state technique, were in the range of $0.2 - 0.6$. Electrochemical impedance spectroscopy analysis of solid polymer electrolytes (SPEs) based on PEO and studied salts with different concentration (from 24 to 94 wt %) was carried out. The ionic conductivity of SPEs was in the order of $10^{-8} - 10^{-7} \text{ S cm}^{-1}$ at room temperature and $10^{-4} \text{ S cm}^{-1}$ at 80°C . A distinguishing feature of SPEs with the studied new salts is the high immobilization of anions, which causes almost a monoconducting character of charge transport. Lithium transference numbers (T_+) exceed 0.9.

Keywords: lithium borate salts; polymer electrolytes; ionic conductivity; lithium transference number; ionic liquids

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1. Introduction

Lithium-ion batteries (LIBs) are, nowadays, the most widely used power sources for electronic devices for which requirements are continuously increasing in terms of their energy and power density, reliability and cyclability [1,2]. Commercial LIBs as electrolyte employ liquid solutions of LiPF_6 in aliphatic carbonates. These solutions are either introduced into porous separator or they are part of a polymer gel [3,4]. LiPF_6 is characterized by relatively high ionic conductivity and is not corrosive against current collectors, but it is known to be thermally and hydrolytically unstable. Its decomposition results in the formation of HF, which can then act destructively on the chemical components of the cell [5-8].

Recently, several new lithium salts have been proposed that could replace LiPF_6 . These include a whole class of imide salts [9], orthoborate [10], salts containing a superweak aluminate anion [11], phosphorus salts (including LiPF_6 derivatives) [12,13] or chelate-type salts [14] and, recently proposed, imidazolid salts [15-18]. However, none of these solutions can yet be found commercially. Therefore, studies on new salts are being conducted, in which attention is focused on developing systems with both high diffusion coefficients and high cation transference numbers [19].

The presence of the low-molecular organic solvents in electrolytes causes a serious problem connected with battery operation safety, since they are volatile and flammable. When the temperature inside a device increases, its unsealing can take place. In emergency situations, the solvent vapors may ignite and such cases have already taken place. Due to the required high capacity of the cell, short charging times and large energy densities, there arises an issue associated with efficient removal of considerable amount of generated heat [20]. Therefore, one of the major problems to be solved is the elimination of flammable organic solvents.

Moreover, liquid electrolytes have one more limitation, namely they are characterized by low share of lithium cations in the transport of electric charge. Lithium transference numbers in typical liquid systems consisting of LiPF_6 in PC solutions, as determined by electrochemical polarization or NMR measurements, are between 0.3 - 0.5 [21]. Mobility of anions causes severe concentration gradients of the conducting salts during current flow and limits the cell current.

In contrast to the liquid electrolytes, solid electrolytes (SEs) do not create risk of ignition or leak and are considered a safe alternative to currently used electrolytes. SEs can be divided into two major groups: inorganic solids - either crystalline, glass or glass-ceramic materials and organic solid polymers [22,23]. Inorganic materials offer high mobility and selective transport of lithium cations within their structure, however the major drawback of many inorganic SEs is their low thermodynamic stability and mechanical properties connected with rigidity and fragility [24]. Organic SEs - polymer electrolytes (SPEs) are flexible and can compensate volume changes of electrodes during charging and discharging by elastic and plastic deformation. SPEs combine solid-state properties with the ease of processing plastic materials. On the other hand, the polymer electrolytes (PEs) exhibit too low ionic conductivity for battery operation at room temperature. One of the approaches to increasing ionic conductivity of PEs relies on the addition of low-molecular plasticizer. In order to avoid evaporation of introduced plasticizers, non-volatile ionic liquids (ILs) can be employed for this purpose [25-29]. However, despite that many of ILs show excellent ionic conductivity, both of their ions migrate at the potential gradient. Cations of IL compete with lithium cations in ionic charge transport and cause a drastic decrease in lithium transference number values ($T_+ < 0.1$) [30,31]. Therefore, many ILs are not suitable for application in lithium-ion batteries. This work is dedicated to the obtaining of salts of the properties of ILs, which are characterized by selective charge transfer with lithium cations. One method of obtaining low-

melting lithium salts is to introduce organic oligomeric chains into the structure of anions. Due to their flexible segmental motion the crystallization is inhibited. Salts have been reported with polyether chains bonded to carboxylic [32], sulfonate [33-35], sulfonamide [36-39] or sulfonimide moieties [40]. Carboxylic acids with oligooxyethylene chains are commercially available. It is easy to obtain salts from them with a simple method, such as reaction with butyllithium. However, the bulk ionic conductivity of carboxylate salts is very low because of poor dissociation of the carboxylates. Our previous works show that the effect of increasing the degree of dissociation can be achieved by complexing anions of the salt by BF_3 [41]. The mechanism of the reactions of BF_3 with carboxylic salts depends on the structure of the anions [42]. The goal of this study is to investigate the structure and properties of products of reactions of lithium carboxylate salts containing oligooxyethylene chains with boron fluoride. Because ionic conductivity is a function of mobility of ions and the number of carrier ions, we selected dicarboxylic salts. We prepared salts of different molecular weights to demonstrate the impact of anion's structure on its mobility and salt's ability to crystallize.

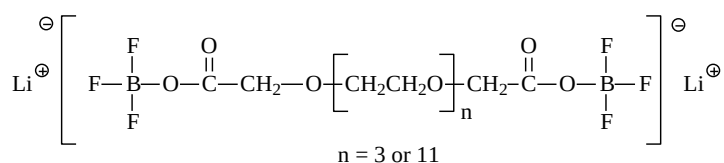
2. Experimental

2.1. Reagents

Prior to use, hexane (anhydrous, 95%, Sigma-Aldrich), tetrahydrofuran (anhydrous, $\geq 99.9\%$, Sigma-Aldrich), acetonitrile (anhydrous, 99.8%, Sigma-Aldrich) and PC (anhydrous, 99.7%, Sigma Aldrich) were dried according to standard methods and stored over molecular sieves in argon atmosphere. Boron fluoride in the form of a diethyl ether complex (Aldrich), n-butyllithium solution (2.5 M in hexanes, Aldrich) and poly(ethylene glycol) bis(carboxymethyl) ethers of various molecular weights ($M_w = 250$ and 600 g mol^{-1} , Aldrich) were used as received. Poly(ethylene oxide) (PEO) ($M_w = 5 \times 10^6 \text{ g mol}^{-1}$, Aldrich) was dried under vacuum for 48 h at 50°C and stored in argon atmosphere.

2.2. Synthesis and characterization of bis(carboxytrifluoroborate lithium) salts $[R(\text{CH}_2\text{COOBF}_3\text{Li})_2]$ with $R=\text{O}(\text{CH}_2\text{CH}_2\text{O})_n$

Bis(carboxytrifluoroborate lithium) salts $[\text{R}(\text{COOBF}_3\text{Li})_2]$ with $\text{R} = \text{O}(\text{CH}_2\text{CH}_2\text{O})_n$ where $n = 3$ or 11] [BCB3 and BCB11] with the following structure:



were obtained in a two-step reaction. The first step involved the reaction between poly(ethylene glycol) bis(carboxymethyl) ethers and *n*-butyllithium. Then, the formed carboxylate lithium salt was reacted with boron fluoride etherate. The course of these reactions is presented in Scheme 1. Both reactions were carried out in dry THF, in the presence of an equimolar amount of substrates. In the first reaction step, *n*-butyllithium was added dropwise to the flask, which required cooling of the system. The second step of the reaction was conducted at room temperature. All operations were carried out under argon atmosphere or on a vacuum line. After the reaction, the solvents formed a separate liquid layer, which was isolated using a syringe and the salts were washed with hexane. The residues of solvents were removed by distillation under reduced pressure. BCB3 was recrystallized from a mixture of acetonitrile and THF, prior to further studies. Depending on the molecular weight of the anion, the obtained salts have different physical properties. The salt with a molecular weight of 400 g mol^{-1} (BCB3) is a crystalline solid, whereas with a molecular weight of 750 g mol^{-1} (BCB11) has the properties of ionic liquid. The structure of the products was determined by ^1H , ^{11}B and ^{19}F NMR spectroscopy.

Scheme 1.

2.3. Preparation of electrolytes

To obtain the polymer electrolytes, the synthesized lithium salts were dissolved in acetonitrile and then mixed with the solution of PEO in acetonitrile. The components formed homogeneous solutions. The solutions were poured onto a poly(tetrafluoroethylene) dish and electrolytes containing from 24 to 94 wt % of salt were formed after vacuum drying. All operations connected with the synthesis of the salts, obtaining of the electrolytes and performing the measurements were carried out in an atmosphere of dried argon. Syringes were used to transport liquids and solid components were weighed in a dry-box (humidity below 50 ppm).

2.4. Measurement techniques

2.4.1. Thermal analysis

Differential scanning calorimetry (DSC) measurements were carried out with a TA Q200 calorimeter under a nitrogen flow. In a dry-box, about 10 mg of salt or electrolyte was sealed in hermetic pans for volatile samples. The measurements were conducted from -100 to 300 °C with a heating/cooling rate of 10 °C min^{-1} .

2.4.2. ^1H , ^{11}B and ^{19}F NMR studies

The structures of the bis(carboxytrifluoroborate lithium) salts were determined by combining ^1H , ^{11}B and ^{19}F NMR spectroscopies. Spectra were recorded on a Varian Mercury 500 spectrometer in CD_3CN at 25 °C. Boron resonance measurements (referenced to $\text{BF}_3\cdot\text{EtO}_2$; 0.0 ppm in CDCl_3) were carried out in quartz tubes. The chemical shifts in the ^{19}F NMR spectra were determined with respect to CF_3COOH .

2.4.3. FTIR spectroscopy

FTIR spectra were obtained by 16 scans with a resolution of 0.1 cm on a Nicolet 6700 FTIR spectrometer with the use of the ATR technique.

2.4.4. Electrochemical measurements

Cyclic voltammetry of bis(carboxytrifluoroborate lithium) salts in PC solution was performed using a VMP3 potentiostat/galvanostat (Bio-Logic, France). Measurements were conducted with a scan rate of 25 mV s⁻¹, at room temperature, where glassy carbon was employed as the working electrode, lithium metal as counter electrode and silver wire as reference.

Ionic conductivity of the electrolytes was determined from impedance measurements using a VMP3 potentiostat/galvanostat (Bio-Logic, France) in the frequency range from 1 Hz to 500 kHz and the temperature range from 20 to 80 °C. The samples were stored in symmetrical cells with stainless steel electrodes. Ionic conductivity was calculated according to the equation:

$$\sigma = R \cdot d/A$$

where R is the resistance determined from the impedance measurement, A is the surface area of the electrode and d is the thickness of the electrolyte.

Lithium transference numbers were determined using the Bruce and Vincent method according to which [43]:

$$T_+ = \frac{I_s(\Delta V - I_0 R_0)}{I_0(\Delta V - I_s R_s)}$$

where ΔV is the potential applied to the sample during polarization [V], I_0 the initial polarization current [A], I_s the steady-state current [A], R_0 the initial interface resistance [Ω] and R_s the steady-state interface resistance [Ω].

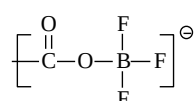
Lithium transference numbers were measured at 60 °C. The samples were placed in symmetrical two-electrode cells between lithium electrodes. For each electrolyte a minimum of three samples were assembled. Interface resistance was determined from the impedance measurement with a variable current signal of 10 mV amplitude in the frequency range from 100 mHz to 500 kHz. Polarization was performed as a chronoamperometric measurement of

20 mV voltage with measurements every 2 s. Polarization was completed when current measured 10 minutes earlier differed by less than 1% from the present one.

3. Results and discussion

3.1. Characterization of bis(carboxytrifluoroborate lithium) salts

For the synthesis two salts were chosen (BCB3 and BCB11). They differ in the number of oxyethylene repeating units, which connect two carboxylate groups: $[R(\text{CH}_2\text{COOBF}_3\text{Li})_2]$ with $R=\text{O}(\text{CH}_2\text{CH}_2\text{O})_n$, for which the average value of $n = 3$ or 11. In the reaction with boron fluoride salts were obtained at high yield (85 - 95%). The structure of the products' anions had been modified and based on the spectral analysis, we identified it as carboxytrifluoroborate:



BCB3 is a crystalline solid with a melting point $T_m = 197.4^\circ\text{C}$ and glass transition temperature $T_g = 27.0^\circ\text{C}$. Fig. 1 shows the thermogram of the salt, which demonstrates that this salt is thermally stable at least up to 250°C . The second salt, which contains approximately 11 repeating units of EO is in the form of a viscous liquid having $T_g = -16.1^\circ\text{C}$.

Figure 1.

Fig. 2 shows FTIR spectrum of the salt obtained in the reaction of lithium carboxylate with boron fluoride (BCB3). Bands of stretching vibrations of the carbonyl group change in frequency from $\nu_{\text{C=O}} = 1600\text{ cm}^{-1}$, which is characteristic for the carboxylate salt $-\text{C}(\text{O})\text{OLi}$, to 1660 and 1700 cm^{-1} for the crystalline salt with $n = 3$, and 1700 cm^{-1} for the liquid salt with $n = 11$. This results from the formation of a covalent bond between an oxygen atom and a boron atom and disturbance of the carboxylic anion hybrid structure, in which carbon is

attached to each oxygen with a strength intermediate between that of C=O and C–O bonds. Frequency of the asymmetric C=O stretching vibration in these new boron anions is lower than in esters and similar as in carboxylic acids. In the crystalline product we observe two bands. This may be due to the presence of two conformers, but certain assignment of these bands requires more spectroscopic studies.

Figure 2.

In ^1H , ^{11}B and ^{19}F NMR spectra, similarly as in FTIR spectroscopy, crystalline salt BCB3 has two groups of signals characteristic for the salt. The B-F groups have signals with chemical shifts in the ^{19}F spectrum at -150.78 and -152.52 ppm, in ^{11}B at 3.90 and 4.43 ppm (Fig. 4). The proportions of these signals vary depending on the polarity of the solvent and for DMSO or D_2O one of the signals disappears. We interpret this behavior by the interaction of the anion with lithium cations: the less polar solvent, the greater share of lithium ions closely related with anions (Scheme 2):

Scheme 2.

The liquid salt BCB11 exhibits one type of chemical environment. BCB11, due to the presence of long, flexible oxyethylene chains, offers greater opportunities for coordination of lithium cations (structure C), which results in a decrease in the share of structure A (Scheme 3).

Scheme 3.

Fig. 3 shows ^{11}B and ^{19}F NMR spectra of BCB11. The main signal in the boron resonance is present at 4.12 ppm and in ^{19}F NMR spectrum at 152.36 ppm and is split by boron isotopes.

Figure 3.

Figure 4.

The presence of oxyethylene groups per molecule of salt is relevant in view of increasing the mobility of the cations, whereas the purpose of substitution of carboxylate group with

boron fluoride is to obtain the effect of ion pair separation. The question arises as to how strongly boron fluoride bonds with the anion and what factors can reverse this reaction. We have examined the influence of nucleophiles of varying strength, such as diglyme, diethylene glycol, acetonitrile, DMSO, H₂O and n-butyllithium on the stability of the bond in the carboxyborate group ($-\text{C}(\text{O})\text{OBF}_3$) in both studied salts. An analysis of changes in FTIR spectra showed that prolonged contact with these solvents, except water, does not cause changes in the salt anions, even at elevated temperature (70 °C). However, in the presence of water gradual hydrolysis occurs. This process leads to the regeneration of the starting carboxylic acid and the formation of new salt – lithium trifluorohydroxyborate:

Scheme 4.

The influence of a strong nucleophile was examined by reaction with n-butyllithium. In this reaction, a much stronger base than the carboxylate anion attacks boron resulting in the release of BF₃ and the formation of new salts:

Scheme 5.

In summary, the interaction with bases stronger than the carboxylate anion, without mobile hydrogen atoms may lead to cleavage of the bond between oxygen and boron causing the formation of a new carboxylate salt and new borate derivative. The presence of the acidic hydrogens lead to the carboxylic acid and lithium borate salt.

Electrochemical investigations of the studied salts were conducted. Fig. 5 presents voltammograms of BCB3 (a) and BCB11 (b). The electrochemical stability of both salts was similar and ranged from –2 to 2 V. In this potential range no oxidation/reduction reaction was observed.

Figure 5.

3.3. Characterization of the electrolytes

In order to characterize the transport properties of the obtained salts, BCB3 and BCB11, their lithium transference numbers (T_+) were determined in solutions in aprotic polar solvents: bis(2-methoxyethyl) ether (diglyme) and propylene carbonate (PC). The obtained values of T_+ are presented in Table 1.

Table 1

As can be seen, the resulting T_+ values for solutions in PC were not high (in the 0.22 – 0.34 range). Despite the relatively high molecular mass of anion salts, their mobility in diluted solutions did not decrease with respect to that of typical inorganic salts. When seeking an explanation of such a behavior it should be noted that the solvent did not interact with the salt anions, but solvated lithium cations. Spectral studies indicate that the coordination number of a lithium ion in PC was 4 – 5, which means that the molecular mass of the lithium cations associated with the solvation shell was equal to ca. 400 – 500 [44]. Therefore, it was greater than that of BCB3 salt anions and slightly lower than that of BCB11. Moreover, oxyethylene monomeric units present in the salt anions participated in the complexation of lithium cations. This interaction, in agreement with the assumptions, is valid from the point of view of salt dissociation, but as a consequence, it limits the mobility of cations. The longer the ether linkage, the greater its share in complexation and this was probably the reason for which the BCB11 salt showed a lower T_+ value (0.224) than that of the BCB3 salt ($T_+ = 0.343$).

A different situation was observed for solutions in diglyme, for which the T_+ value increased up to 0.6. This solvent competed stronger with BCB3 anions in complexing lithium ions. The smaller number of solvent molecules in the solvation shell can explain the greater mobility of Li^+ cations in this solvent.

~~As can be seen, the resulting T_+ values were high, above 0.6, which means that they are substantially higher than those obtained with inorganic salts often studied [21]. Lower value~~

~~for the solution in diglyme indicated that this solvent provided worse conditions for the transport of Li^+ . The cause of this effect may be the formation of the solvation shell from diglyme, which significantly reduces the mobility of cations. The results obtained for PC does not give answers why lithium transference number for large anions of BCB11 were lower than for BCB3.~~

The obtained salts were used as components of solid polymer electrolytes with PEO as a polymer matrix. Membranes contained different concentrations of salts, expressed by the ratio of repeating units of lithium cations to ethylene oxide (EO), from 1:15 to 2:1 (Li:EO).

The properties of the membranes depended on the content of salt and its molar mass. The content of crystalline phase and phase transition temperatures were determined from DSC measurements and are presented in Table 2.

Table 2

Fig. 6 shows a photograph of a membrane comprising 82 wt% of BCB3 (Li:EO equal 1:1), in which we can see that the membrane is practically amorphous while maintaining good mechanical properties. Thermogram of this electrolyte does not contain the peak characteristic for melting of the crystalline phase of PEO.

Figure 6.

At lower salt concentrations, for Li:EO ratio equal 1:15 and 1:9, PEO forms a crystalline phase in the presence of both salts. Fig. 7 shows the DSC curves for the polymer electrolytes having the ratio of Li:EO ratio 1:15, to illustrate the effect of BCB3 and BCB11 on the crystallization of the polymer matrix.

The data in the Table 2 indicate that the addition of the salt with smaller anion stronger inhibits the crystallization of PEO and at high concentrations of this salt results in completely amorphous electrolytes. In case of BCB11 with long oxyethylene segments, the calculated

content of PEO crystalline phase is higher. A small thermal effect associated with the melting of PEO crystalline phase is present even at molar excess of the salt.

Figure 7.

For polymer membranes containing BCB3 and BCB11 salts, ionic conductivities were measured, the results of which are shown in Fig. 8. The ionic conductivity values and changes as a function of temperature for 1:15 and 1:9 Li:EO ratio are typical like for crystalline electrolytes based on PEO [45-47]. The room temperature values of conductivity are in the order of 10^{-8} S cm $^{-1}$ and increase to 10^{-4} S cm $^{-1}$ at 80 °C. At high concentrations of salt, for the ratio equal 1:3 and higher, electrolytes with BCB11 are characterized by higher conductivity in the order of 10^{-7} S cm $^{-1}$ at RT and 10^{-4} S cm $^{-1}$ at 80 °C. This dependence is consistent with the VFT equation, which is characteristic for amorphous electrolytes.

Below the melting point of the crystalline phase of PEO, better conductivity exhibit systems containing high concentrations of lithium salts. At Li:EO ratio equal 1:3 and higher (over 50 wt% of salt) the composition of electrolytes is characteristic for so-called polymer-in-salt materials. There are some suggestions that in SPE of high salt concentration, separated single clusters come into contact with each other, thus forming an infinitive cluster promoting fast ionic transport [48]. Above the melting point of PEO the electrolytes with a low salt concentration show higher conductivity. Probably, at elevated temperature, the charge transport occurring by the segmental motion of the polymer chains is more effective than that due to rearrangements within the ionic clusters.

Figure 8.

The conductivity of the analyzed systems is relatively low, but if it was essentially due to the movement of lithium cations, then such systems would have advantage over other SPEs

based on PEO complexes. To explain the nature of conductivity, lithium transference numbers of studied polymer electrolytes were measured. The results are presented in Table 3.

Table 3

As can be noticed, SPE in the presence of the studied salts were characterized by much greater T_+ values with respect to that of solutions in PC. For BCB3 the ion transport conditions are comparable to that of solutions in diglyme (T_+ is ca. 0.6). An increase in the anion molecular mass in BCB11 caused a considerable decrease in their mobility when high molecular weight PEO is the solvent for the salt ($T_+ > 0.9$). The lithium ions transport proceeds *via* the chain segmental movements without the solvation shell, due to which they become more mobile than anions. Anions undergo immobilization by tangling with PEO chains and by interaction with Li^+ they are trapped in the complex: anion-- Li^+ --PEO. It can be assumed that the electrolytes involving BCB11 are polymer electrolytes plasticized by an ionic liquid, with very specific for those systems monoconducting properties.

4. Conclusions

The results of the presented studies show that the investigated method of synthesis of new bis(carboxytrifluoroborate lithium) salts is effective in boron fluoride chemical bonding with carboxylate anions. This bond is stable in the presence of nucleophiles of medium strength such as ethers, nitriles or sulfones. The reaction with butyllithium caused the release of BF_3 from carboxytrifluoroborate anions and led to the formation of carboxylate salt and lithium butyltrifluoroborate. Studies on hydrolytic stability showed the formation of carboxylic acid and lithium hydroxytrifluoroborate salt.

New bis(carboxytrifluoroborate lithium) salts with oligooxyethylene chains were crystalline solids or had properties of room temperature ionic liquids thermally stable to 250

°C. They exhibited lithium transference numbers comparable to those of inorganic lithium salts like LiPF₆ in diluted solutions in PC ($T_+ = 0.2 - 0.3$).

Solid polymer electrolytes based on PEO and studied salts were characterized by conductivity in the order of $10^{-8} - 10^{-7}$ S cm⁻¹ at room temperature and 10^{-4} S cm⁻¹ at 80 °C. The distinguishing feature of electrolytes with these new salts is high immobilization of anions, which causes almost a monoconducting character of charge transport. Lithium transference numbers (T_+) exceed 0.9.

New studied salts, both as ionic liquids and as part of the polymer systems, seem to be promising electrolytes, although further research is needed in terms of improving their ionic conductivity.

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Table 1.

Lithium transference numbers of BCB3 and BCB11 0.1M salts solutions in diglyme and PC.

Salt	Solvent	R_0 [Ω]	R_s [Ω]	I_0 [A]	I_s [A]	T_+	Average T_+
BCB3	diglyme	591	998	0.026	0.016	0.707	0.600
BCB3	diglyme	481	749	0.037	0.023	0.494	
BCB3	PC	52	59	0.273	0.159	0.318	0.343
BCB3	PC	55	59	0.211	0.139	0.469	
BCB3	PC	487	613	0.035	0.023	0.335	
BCB3	PC	400	495	0.044	0.029	0.272	
BCB3	PC	274	388	0.055	0.030	0.323	
BCB11	PC	224	307	0.035	0.005	0.094	0.224
BCB11	PC	99,6	142	0.091	0.031	0.239	
BCB11	PC	198	234	0.068	0.031	0.233	
BCB11	PC	218	257	0.070	0.034	0.201	

Table 2.

DSC data of polymer electrolytes containing PEO and BCB3 and BCB11salts with different salt:PEO ratios.

Li:EO BCB3				Li:EO BCB11			
	T _g ^a (°C)	T _m ^b (°C)	X _c ^c (%)		T _g ^a (°C)	T _m ^b (°C)	X _c ^c (%)
1:15	-17.5	65.7	52	1:15	-21.8	69.9	57
1:9	-41.3	60.0	28	1:9	-17.9	62.0	47
1:3	-45.3	-	-	1:3	-16.9	56.4	20
1:1	-64.3	-	-	1:1	-19.9	54.9	5
2:1	-69.1	-	-	2:1	-18.0	53.5	2.5

^a Determined from 2nd heating cycle

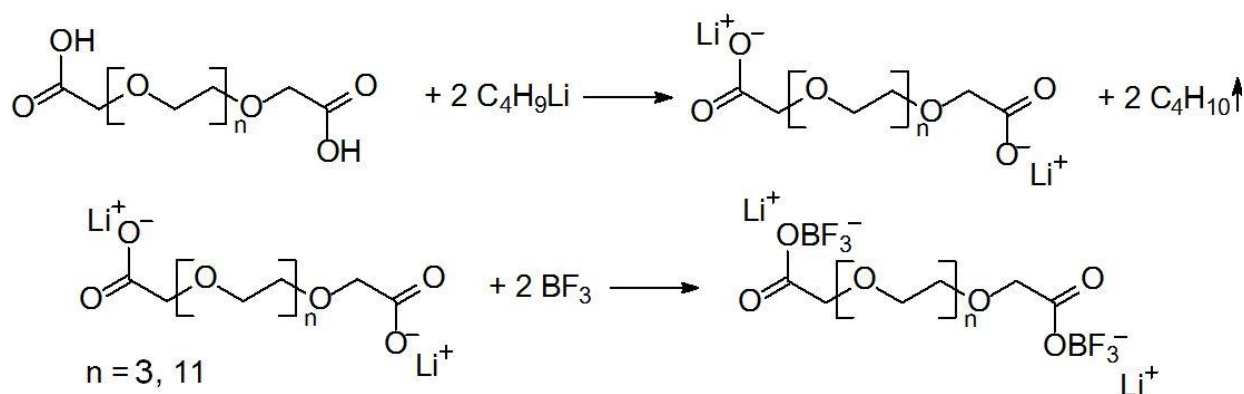
^b Determined from 1st heating cycle

^c Degree of crystallinity of PEO calculated with respect to PEO content in electrolytes

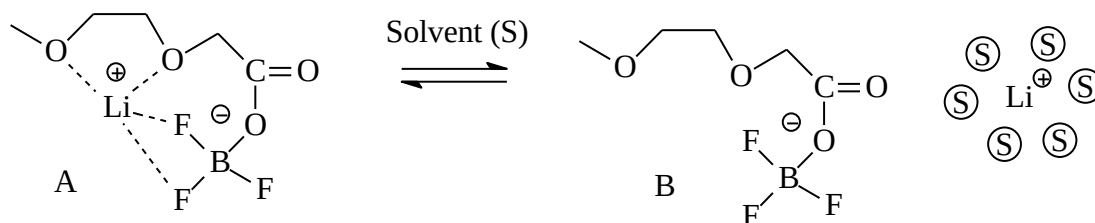
Table 3.

Lithium transference numbers of polymer electrolytes containing PEO and BCB3 or BCB11 at various molar ratio of Li:EO.

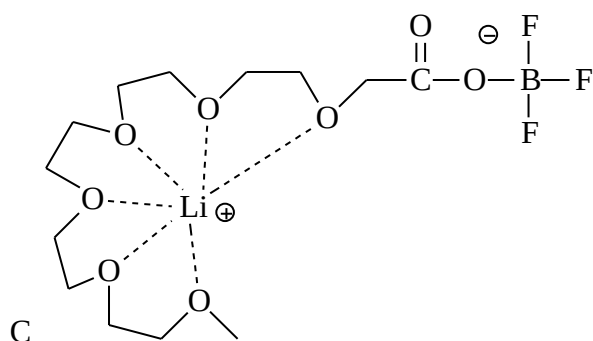
Salt:	BCB3		BCB11		
wt % of salt	33	82	36	49	74
Li:EO	1 : 9	1 : 1	1 : 15	1 : 9	1 : 3
T ₊	0.58	0.60	0.86	0.94	0.94



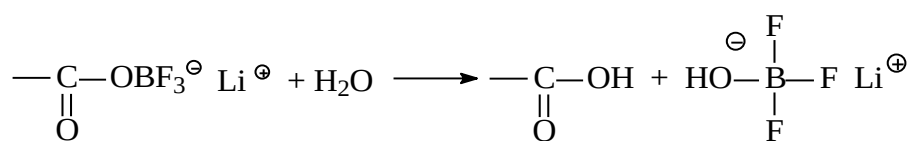
Scheme 1. Synthesis of bis(carboxytrifluoroborate lithium) salts in the reaction of poly(ethylene glycol) bis(carboxymethyl) ethers with n-butyllithium and boron fluoride etherate.



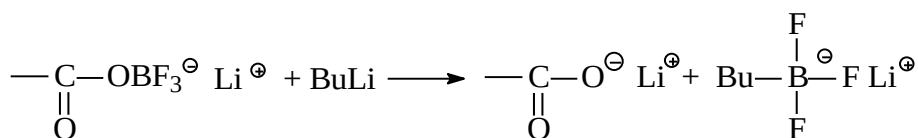
Scheme 2. Effect of polar solvent on the mode of coordination of lithium cations in BCB3 salt.



Scheme 3. The mode coordination of lithium cations in BCB11 salt.



Scheme 4. Reaction of hydrolysis of lithium carboxytrifluoroborate salts.



Scheme 5. Reaction of lithium carboxytrifluoroborate salts with butyllithium.

Captions to Figures

Fig. 1. Thermogram of bis(carboxytrifluoroborate lithium) salt: BCB3 (the first heating cycle).

Fig. 2. FTIR spectrum of BCB3 performed with ATR technique.

Fig. 3. ^{11}B and ^{19}F NMR spectra of BCB11 in CD_3CN .

Fig. 4. ^{11}B and ^{19}F NMR spectra of BCB3 in CD_3CN .

Fig. 5. Voltammograms of the bis(carboxytrifluoroborate lithium) salt BCB3 (a) and BCB11 (b) in 0.1 M PC solution.

Fig. 6. Picture of polymer electrolyte membrane containing PEO and 82 wt% of BCB3 (Li:EO equal 1:1).

Fig. 7. Thermograms of polymer electrolytes containing PEO with the addition of BCB3 and BCB11 salts at molar ratio of Li:EO equal 1:15.

Fig. 8. Ionic conductivity of polymer electrolytes containing PEO with salts: BCB3 (a) and BCB11 (b) at various molar ratio of Li:EO.

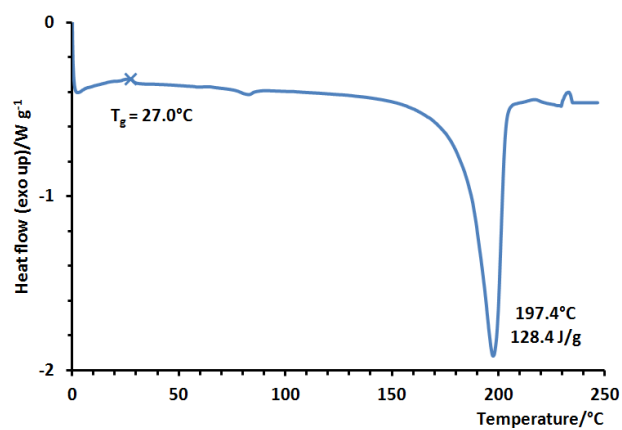


Fig. 1. Thermogram of bis(carboxytrifluoroborate lithium) salt: BCB3 (the first heating cycle).

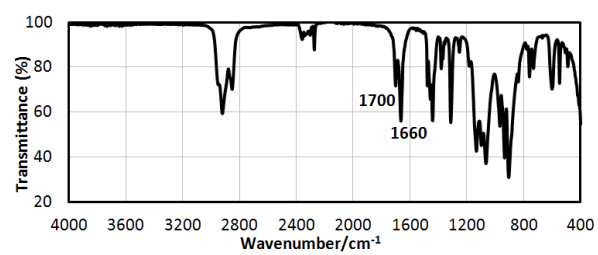


Fig. 2. FTIR spectrum of BCB3 performed with ATR technique.

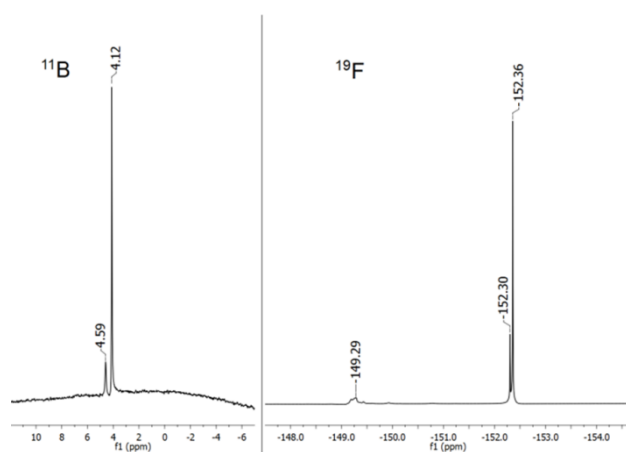


Fig. 3. ^{11}B and ^{19}F NMR spectra of BCB11 in CD_3CN .

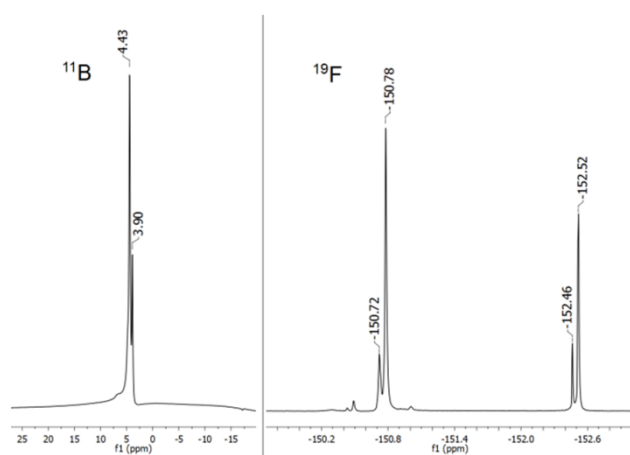


Fig. 4. ^{11}B and ^{19}F NMR spectra of BCB3 in CD_3CN .

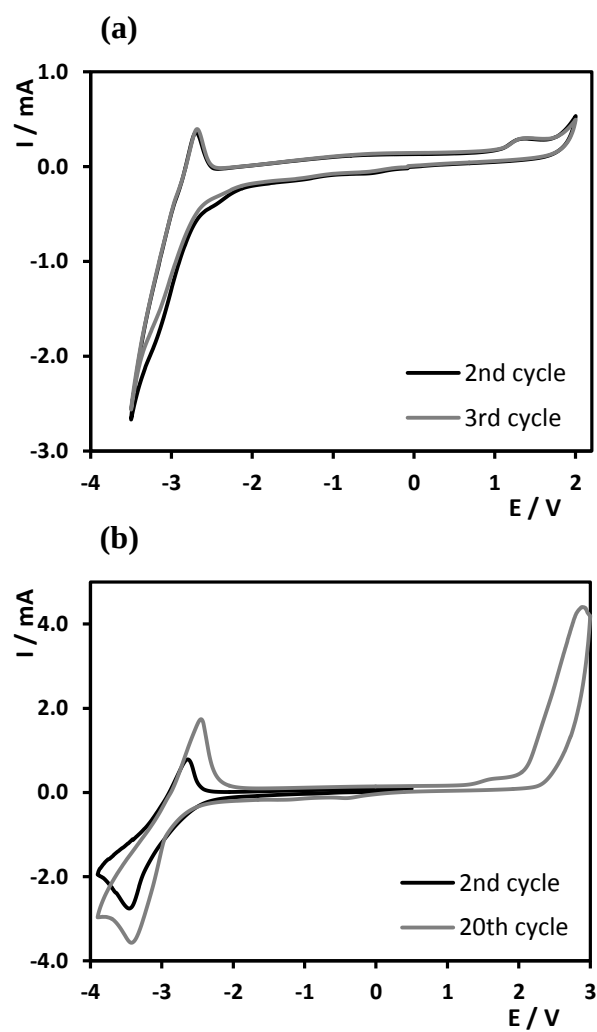


Fig. 5. Voltammograms of the bis(carboxytrifluoroborate lithium) salt BCB3 (a) and BCB11 (b) in 0.1 M PC solution.



Fig. 6. Picture of polymer electrolyte membrane containing PEO and 82 wt% of BCB3 (Li:EO equal 1:1).

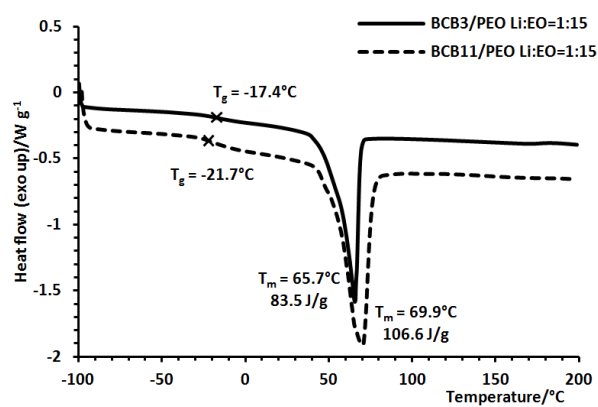


Fig. 7. Thermograms of polymer electrolytes containing PEO with the addition of BCB3 and BCB11salts at molar ratio of Li:EO equal 1:15.

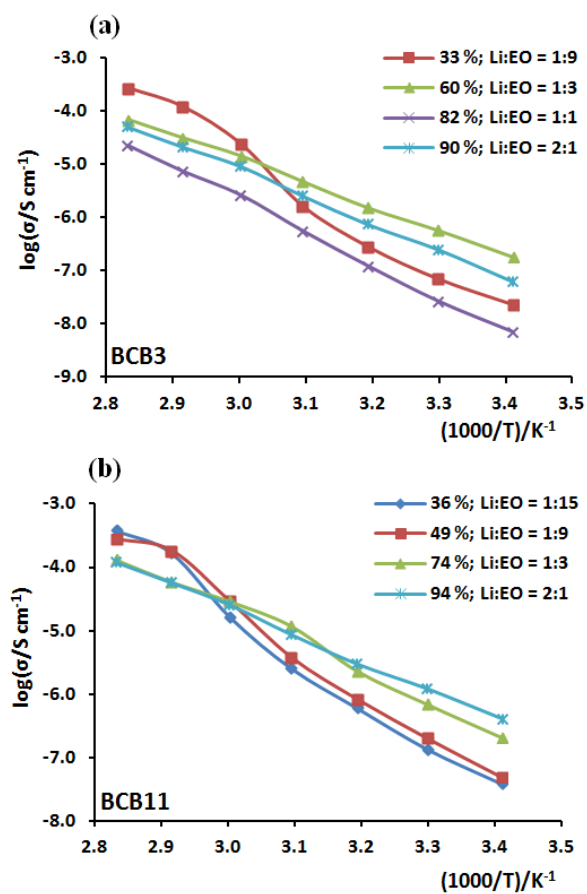


Fig. 8. Ionic conductivity of polymer electrolytes containing PEO with salts: BCB3 (a) and BCB11 (b) at various molar ratio of Li:EO.